Complete k2 row combination: unreviewed proof packet

UNREVIEWED complete k2: two distinct statements

Raw pivot, a nonzero: proposed equality of highest-part regularity

Add N1+2aN2, b nonzero: proposed strict decrease of at least one

All auxiliary rows retained; A0 alone eliminated; q remains projected

Complete k2: raw equality and added-row strictness

UNREVIEWED producer drafts. The two frozen derivations are preserved verbatim below. The raw substitution draft proposes equality of highest-part regularity for nonzero target x. The added-row draft proposes a strict drop of at least one for nonzero target y after appending N1+2a N2 and eliminating A0. Both concern the specified complete finite k2 constant-B chart with every auxiliary and field constraint retained, under their exact nonzero/finite quotient premises. Independent derivation is pending.

The runtime transformation eliminates only A0; q=A1 remains projected. The graded quotient calculation is a proof step. This packet does not assert solving-degree or SAT-cost improvement. The original 106 coordinate and separate six coefficient cohorts, complete projected outputs, reconstruction, failed work and certificate costs remain requirements. No candidate import, suite, CAS, solver, census or scientific run was executed.

UNREVIEWED SOURCE-BOUND DERIVATION — raw k2 pivot can preserve regularity exactly

Scope. Handwritten algebra and direct source inspection only. This is a new argument outside DEC-20261102-cd6ac3, not part of its sealed independent review. No code import, matrix, solver, CAS or census was run. The statement concerns the literal original-X norm-coefficient rows in canonical_chart, followed solely by retained-row affine substitution, with optional certified deletion of the eliminated field row. It does not concern target-centered row recombination, Boolean compilation or solving degree.

Source normalization. For finite k=2, chart_spec has d=3, c=−1 and ordered basis (A0,A1,B0), with B0 normalized to 1. Thus A=A0+A1 X, B=1, U=X²+u1 X+u0, and curve f=X³+alpha X+beta. For public target (a,b), b²=a³+alpha a+beta, the norm is A²−f+(X−a)U. Its nonzero coefficient rows are N2=A1²+u1−a, N1=2A0 A1+u0−a u1−alpha, N0=A0²−a u0−beta. The sign row is L=A0+a A1−b. The cubic coefficient is identically zero.

Admissible stratum. In the constant stratum, B1=0 and B0=1; the inverse row is z−1. Interval rows depend only on U and interval auxiliaries, never A0. The uniform nilpotent arm imports a component bound only to U. The other two arms in the constant stratum import the constant component bound only to B0=1. Therefore every imported rationality row is independent of A0 in all three arms. This conclusion uses the binding loop, not execution of the component builder. In the zero stratum B0=0 is the scalar row 1; in the linear stratum B1*z−1 is the scalar row −1. Both are literal unit-ideal cases before and after substitution, regardless of other rows.

Claim for a≠0. Let R contain all original chart and auxiliary variables; form F from the literal full canonical rows plus every variable field equation v^p−v, with p>3. Let G result from A0=b−a A1, retaining every nonzero substituted row. Then the original and reduced highest-part quotients are graded-isomorphic. In particular their degree-of-regularity values are EQUAL for every allowed interval and every rationality arm, for both b=0 and b≠0. No exact value is asserted.

Proof. Let phi be homogeneous A0↦−a A1, with all other variables unchanged. The pivot's highest part generates ker(phi). The norm highest parts map respectively to A1², −2a A1² and a² A1². Since p>3 and a≠0, all three images are nonzero, so the affine-substituted norm rows retain degree two and have exactly these highest parts. Every other nonzero non-pivot row in the constant stratum is independent of A0 and is unchanged, except the A0 field row. That field row becomes (b−a A1)^p−(b−a A1)=−a(A1^p−A1), with nonzero highest part −a A1^p. All other field rows are unchanged. Therefore the reduced highest ideal J is exactly phi(I), where I is the original highest ideal; no extra lower-degree highest generator is exposed. Since ker(phi)⊆I, R/I≅S/phi(I)=S/J as graded rings. Dropping the transformed A0 field row preserves J because the A1 field row remains. All-variable field augmentation supplies finiteness. In the two impossible strata both quotients are zero, so equality also holds trivially.

Why a=0 is excluded. After A0=b, N1 becomes 2b A1+u0−alpha, a nonzero linear row; N0 becomes b²−beta=0. The homogeneous image of N1_top is zero, so the preceding equality proof fails. This creates a possible strictness mechanism, not a proof that the complete chart is strict. The actual auxiliary rows must still be included in that analysis.

Why the translated diagnostic does not contradict this. Translation of the polynomial variable X to Z=X−a also recombines norm COEFFICIENT ROWS. In original coordinates, the Z coefficient row is N1+2a N2. After the pivot it becomes u0+a u1+2b A1−alpha−2a², which is linear. Before substitution it still has a quadratic highest part 2A1(A0+a A1). Although this row is an ordinary linear combination of retained norm rows, its newly exposed highest part need not belong to the ideal generated by their individual post-substitution highest parts. Thus target-centered coefficient recombination is a material presentation change, not merely relabeling the variables. The reviewed diagnostic performs that change; a literal raw-row substitution does not.

Construction implication, unreviewed. To pursue strictness for general nonzero target abscissae, an implementation should consider explicit target-centered coefficient recombination or certified addition of its exposed linear consequence, then retain that row. The later two-pivot draft uses precisely this type of extra consequence, but its complete-chart theorem and costs are not accepted here. Compare exact original projections and charge every transformation and certificate. The original 106/6 cohorts and all five leads remain required; this argument authorizes no scientific run and does not establish a timing result.

UNREVIEWED CONSTRUCTION — a strict complete-k2 reduction from one norm-row combination

Status and scope. This is a new handwritten proof candidate, not covered by the sealed affine review or DEC-20261102-cd6ac3. It concerns the literal complete finite k=2 chart, every retained interval and rationality constraint, the constant B stratum, all three existing rationality arms, and explicit field equations for EVERY variable. No implementation, solver, CAS, matrix computation or census was run. The prime is p>3, target T=(a,b) is on $y^2=X^3+\text{alpha } X+\text{beta}$, and b≠0. Assume the original highest-part quotient is nonzero; all-variable field equations ensure it is finite. Impossible strata with scalar obstructions have regularity zero and are excluded from any strictness claim.

Actual source structure. Finite k=2 normalizes B=1 and has A=A0+qX and U=X²+u1X+u0. The norm rows are N2=q²+u1−a, N1=2A0q+u0−a u1−alpha, and N0=A0²−a u0−beta; the sign row is L=A0+a q−b. In the constant stratum all other rows use only support, inverse, interval and rationality variables, denoted collectively z. For the uniform arm, rationality is bound only to U; for the other two arms it is bound only to B0=1. Thus none of those other rows uses A0 or q. Keep their exact nonzero highest parts and every z field-row highest part in an ideal K⊂F_p[z]. The original rows themselves remain untouched.

Original quotient, both inclusions. Let I be the highest-part ideal of the full original system in F_p[A0,q,z]. Its norm/sign generators are q², 2A0q, A0², A0+a q, and the coefficient fields add A0^p,q^p. All other generators belong to K. The ideal (A0+a q,q²,K) contains 2A0q and A0² by reduction modulo A0+a q and q²; it contains A0^p and q^p since p>3. Conversely A0+a q and q² are actual highest parts of L and N2, and every generator of K is an actual unchanged row's highest part. Therefore I=(A0+a q,q²,K). Set C=F_p[z]/K. This gives a graded isomorphism R/I≅C[q]/(q²), with no assumption on the complicated internal relations of C. If C has largest nonzero degree d, the original quotient has largest degree d+1: q times any nonzero degree-d class survives. Hence dreg(F)=d+2. C is finite and nonzero under the stated premises.

The added equation and its cost boundary. Add the single row R=N1+2a N2 before substituting the pivot. This is a constant-coefficient linear combination of existing equations, so it changes neither the ordinary ideal nor the solution set. Its highest part is 2q(A0+a q), which is already in I. Hence its addition also leaves the original highest-part ideal and degree of regularity unchanged. For a=0 this row duplicates N1; duplicate removal is harmless. Constructing and checking this relation must be charged in a future implementation; no computational cost measurement is claimed here.

After substitution. Set A0=b−a q and retain ALL nonzero substituted rows, including the new row. The new row becomes g=u0+a u1+2bq−alpha−2a². Let v=u0+a u1. Its highest part is v+2bq, with nonzero coefficient 2b on q. N2 retains highest part q². If a≠0, N1 and N0 retain respective highest parts −2a q² and a²q². If a=0, N1 is g and N0=b²−beta=0 because T is on the curve. All rows contributing K remain unchanged. The eliminated coefficient field row is −a(q^p−q), or zero when a=0; its highest part is redundant modulo q². Thus, in every abscissa case, the reduced highest-part ideal is exactly J=(q²,K,v+2bq). Each displayed generator comes from a retained row, and every retained row's highest part is in the displayed ideal. These are both ideal inclusions, not just one upper containment.

Strictness. Since 2b is invertible, eliminate q linearly in S/J to obtain the graded isomorphism S/J≅C/(v²). Indeed q=−v/(2b), and q² becomes a nonzero scalar multiple of v²; the converse map sends z to z and q to −v/(2b). The largest nonzero degree of C/(v²) is at most d. It follows that dreg(G)≤d+1=dreg(F)−1. This proves a strict decrease of at least ONE in the chosen highest-part regularity invariant, conditional on C≠0. It does not rely on diagnostic thresholds 2p or p+1, on knowing d, or on dropping any auxiliary equation. The reduced quotient is proper when C is proper because v² has positive degree; the zero-quotient convention would only strengthen the upper bound.

Uniformity and exceptions. The argument allows every original t,s and wrapped interval, every auxiliary field equation and all three rationality arms in the constant stratum. It does not require the original equations to have a solution; it requires only a nonzero original highest-part quotient. Any valid full witness implies that premise because no nonzero constant row could then occur, but such a witness is not assumed or enumerated here. Zero and linear B strata contain scalar obstructions under B=1 and cannot have a strict drop below zero. Identity targets and other k are separate presentations. When b=0, the exposed relation v lacks q: the quotient is (C/(v))[q]/(q²), so strictness depends on whether the top component of C survives mod v. No uniform strictness claim is made for b=0.

Why this differs from raw substitution. For a≠0, literal norm rows alone expose no new lower-degree highest part; their quotient remains C[q]/(q²), as in the separate equality draft. Adding R costs an ordinary redundant row but changes which highest parts are exposed AFTER substitution. It is essential to retain that new row. This is the concrete structural modification connecting the raw full chart to target-centered coefficient normalization.

Scope of progress. If independently verified against the complete source bindings, this establishes a strict presentation-level degree-of-regularity reduction for a broad complete k2 family. It is not yet a checked compiler transformation, a solving-degree theorem, or a measured SAT speedup. The transformation preserves full solutions by the added-row identity and affine reconstruction; the separate projected-blocking proof and complete original 106/6 model-set cohorts still need authenticated implementation and full cost accounting. Future work must certify row transport and compare original outputs, with failed reconstruction and certificate work retained.